POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Microprocessor and ALP | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Von Neumann and Harvard architecture. Explain fetch, decode and execute cycle in microprocessor with suitable diagram. 7
   b) What are the different rotate instructions used in 8085 instruction sets? Explain with the help of simple example for each of them. 8
2. a) Draw and explain a well labelled timing diagram of the instruction LXI H, 2435H. 8

   OR

   Draw the timing diagram for the 8085 instruction LDAX D.
   b) Write an 8085 ALP to add the odd numbers in a set of data array. The length of an array is in memory location C050H and starting address of the array is from C051H. Store the sum and Carry in memory location D050 and D051. 7
3. a) Write an assembly language program to display a string 'POKHARA UNIVERSITY' characterwise using macro to insert space between characters. 7

   OR

   Write an assembly language program to display 'MICROPROCESSOR' in reverse order.
   b) Draw the internal architecture of 8086 microprocessor and compare the functions of its two basic units. 8
4. a) Explain detail about the memory segmentation of 8086 microprocessor with its diagram. 7
   b) What is the significance of 8255 PPI? Draw and explain the control word for 8255 PPI in I/O mode. 8

5. a) Differentiate between partial and absolute address decoding. Design memory interfacing circuit with 8085 for 2KB RAM, 4KB ROM and 8KB EPROM with initial address 1000 H. 8
   b) What is DMA? Explain DMA operation with the block diagram of DMA Controller 7
6. a) What are the hardware and software interrupts of 8085 and 8086? Briefly describe the conditions which cause the 8086 to perform each of the following types of interrupts: Type 0, Type 1, Type2, Type 3 and Type 4. 8
   b) Explain the block diagram of Programmable Interrupt Controller with its major features. 7
7. Write short notes on: **(Any two)** 2×5
   a) Stack and Subroutine
   b) Maskable and non-maskable interrupts of 8085
   c) Polled vs. Chained interrupt

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